

## SAFETY DATA SHEET

Creation Date 11-Oct-2011

Revision Date 25-Mar-2024

Revision Number 3

### 1. Identification

**Product Name** Ethylenediaminetetraacetic acid, 0.5M aqueous solution. pH 8.0

**Cat No. :** J60292

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                      |            |
|------------------------------------------------------|------------|
| Serious Eye Damage/Eye Irritation                    | Category 2 |
| Specific target organ toxicity - (repeated exposure) | Category 2 |
| Target Organs - Respiratory system.                  |            |

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

Causes serious eye irritation

May cause damage to organs through prolonged or repeated exposure



## Prevention

Wear eye/face protection

Do not breathe dust/fume/gas/mist/vapors/spray

## Response

Get medical attention/advice if you feel unwell

## Eyes

If eye irritation persists: Get medical advice/attention

## Disposal

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

| Component                               | CAS No    | Weight % |
|-----------------------------------------|-----------|----------|
| Water                                   | 7732-18-5 | > 85     |
| Ethylenediamine tetraacetic acid (EDTA) | 60-00-4   | 10 - 15  |

|                                            |                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                      |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable.                                                                                      |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                             |

|                                       |                          |
|---------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b> | No information available |
|---------------------------------------|--------------------------|

|             |                          |
|-------------|--------------------------|
| Flash Point | Not applicable           |
| Method -    | No information available |

|                                  |                          |
|----------------------------------|--------------------------|
| Autoignition Temperature         | Not applicable           |
| Explosion Limits                 |                          |
| Upper                            | No data available        |
| Lower                            | No data available        |
| Sensitivity to Mechanical Impact | No information available |
| Sensitivity to Static Discharge  | No information available |

#### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

#### Hazardous Combustion Products

None known.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

|        |              |             |                  |
|--------|--------------|-------------|------------------|
| Health | Flammability | Instability | Physical hazards |
| 2      | 0            | 0           | N/A              |

### 6. Accidental release measures

|                           |                                                                             |
|---------------------------|-----------------------------------------------------------------------------|
| Personal Precautions      | Use personal protective equipment as required. Ensure adequate ventilation. |
| Environmental Precautions | Do not flush into surface water or sanitary sewer system.                   |

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### 7. Handling and storage

**Handling** Ensure adequate ventilation. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place.

### 8. Exposure controls / personal protection

**Exposure Guidelines** This product does not contain any hazardous materials with occupational exposure limitsestablished by the region specific regulatory bodies.

**Engineering Measures** Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

#### Personal Protective Equipment

|                                 |                                                                                                                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face Protection</b>      | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.                                                                       |
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                                         |
| <b>Respiratory Protection</b>   | Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. |
| <b>Recommended Filter type:</b> | Particulates filter conforming to EN 143.                                                                                                                                                                                                         |

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Liquid                   |
| Appearance                             | Colorless                |
| Odor                                   | Slight                   |
| Odor Threshold                         | No information available |
| pH                                     | 8                        |
| Melting Point/Range                    | No data available        |
| Boiling Point/Range                    | No information available |
| Flash Point                            | Not applicable           |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | No information available |
| Specific Gravity                       | No information available |
| Solubility                             | miscible                 |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | Not applicable           |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |

## 10. Stability and reactivity

|                                  |                                          |
|----------------------------------|------------------------------------------|
| Reactive Hazard                  | No                                       |
| Stability                        | Stable under normal conditions.          |
| Conditions to Avoid              | Incompatible products.                   |
| Incompatible Materials           | Strong oxidizing agents                  |
| Hazardous Decomposition Products | None under normal use conditions         |
| Hazardous Polymerization         | Hazardous polymerization does not occur. |
| Hazardous Reactions              | None under normal processing.            |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| Oral LD50   | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Dermal LD50 | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Vapor LC50  | Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.    |

#### Component Information

| Component                               | LD50 Oral                                 | LD50 Dermal | LC50 Inhalation |
|-----------------------------------------|-------------------------------------------|-------------|-----------------|
| Water                                   | -                                         | -           | -               |
| Ethylenediamine tetraacetic acid (EDTA) | 4500 mg/kg ( Rat )<br>>2000 mg/kg ( Rat ) | Not listed  | 1 mg/l (rat)    |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|            |                    |
|------------|--------------------|
| Irritation | Irritating to eyes |
|------------|--------------------|

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                               | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------------------------------|-----------|------------|------------|------------|------------|------------|
| Water                                   | 7732-18-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Ethylenediamine tetraacetic acid (EDTA) | 60-00-4   | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known  
**STOT - repeated exposure** Respiratory system

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Contains a substance which is: Toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component                               | Freshwater Algae                                 | Freshwater Fish                                                                                                  | Microtox   | Water Flea                                   |
|-----------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------|
| Ethylenediamine tetraacetic acid (EDTA) | EC50: = 1.01 mg/L, 72h (Desmodesmus subspicatus) | LC50: 34 - 62 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 44.2 - 76.5 mg/L, 96h static (Pimephales promelas) | Not listed | EC50: = 113 mg/L, 48h Static (Daphnia magna) |

**Persistence and Degradability** Miscible with water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT** Not regulated  
**TDG** Not regulated  
**IATA** Not regulated  
**IMDG/IMO** Not regulated

## 15. Regulatory information

### United States of America Inventory

| Component                               | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-----------------------------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Water                                   | 7732-18-5 | X    | ACTIVE                                        | -                           |
| Ethylenediamine tetraacetic acid (EDTA) | 60-00-4   | X    | ACTIVE                                        | -                           |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)** Not applicable

**TSCA 12(b)** - Notices of Export Not applicable

### International Inventories

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                               | CAS No    | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|-----------------------------------------|-----------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Water                                   | 7732-18-5 | X   | -   | 231-791-2 | X     | X    |      | X    | X     | KE-35400 |
| Ethylenediamine tetraacetic acid (EDTA) | 60-00-4   | X   | -   | 200-449-4 | X     | X    | X    | X    | X     | KE-13648 |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

#### SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

#### SARA 311/312 Hazard Categories

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

### CWA (Clean Water Act)

| Component                               | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|-----------------------------------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Ethylenediamine tetraacetic acid (EDTA) | X                          | 5000 lb                     | -                      | -                         |

**Clean Air Act** Not applicable

**OSHA** - Occupational Safety and Health Administration Not applicable

### CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

| Component                               | Hazardous Substances RQs | CERCLA Extremely Hazardous Substances RQs | SARA Reportable Quantity (RQ) |
|-----------------------------------------|--------------------------|-------------------------------------------|-------------------------------|
| Ethylenediamine tetraacetic acid (EDTA) | 5000 lb                  | -                                         | 5000 lb<br>2270 kg            |

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component                               | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-----------------------------------------|---------------|------------|--------------|----------|--------------|
| Water                                   | -             | -          | X            | -        | -            |
| Ethylenediamine tetraacetic acid (EDTA) | X             | X          | X            | -        | -            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH**

| Component                               | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Water                                   | 7732-18-5 | -                                                                   | -                                                                             | -                                                                                                     |
| Ethylenediamine tetraacetic acid (EDTA) | 60-00-4   | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                               | CAS No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------------------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Water                                   | 7732-18-5 | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Ethylenediamine tetraacetic acid (EDTA) | 60-00-4   | Listed   | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component                               | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water                                   | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Ethylenediamine tetraacetic acid (EDTA) | 60-00-4   | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By** Health, Safety and Environmental Department  
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www.thermofisher.com

**Creation Date** 11-Oct-2011  
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**Revision Summary** New emergency telephone response service provider.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**